

GLA UNIVERSITY

Application Build and Release Automation Lab

ASSIGNMENT – 4

Linux and Git Version Control

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Roll Number: 2415001871

Subject Code: BCSE1522

Semester: 5th

Question 1 — Project Workspace

```
Last login: Tue Sep 15 11:52:25 on ttys000
[abhijeetsingh@Abhijeets-MacBook-Air ~ % cd ~/Cloud-Assignment
[abhijeetsingh@Abhijeets-MacBook-Air Cloud-Assignment % mkdir Abhijeet-GLA-LibrarySystem
mkdir: Abhijeet-GLA-LibrarySystem: File exists
[abhijeetsingh@Abhijeets-MacBook-Air Cloud-Assignment % cd Abhijeet-GLA-LibrarySystem
[abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % mkdir src docs config backup
mkdir: src: File exists
mkdir: docs: File exists
mkdir: backup: File exists
[abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % touch library_info.txt
[abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % nano library_info.txt
[abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % pwd
/Users/abhijeetsingh/Cloud-Assignment/Abhijeet-GLA-LibrarySystem
[abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % ls
backup          config          docs            library_info.txt  src
[abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % cat library_info.txt
Name: Abhijeet Singh
Roll Number: 2415001871
Subject: Application Build and Release Automation
Project Name: Online Library Management System
Semester: 5th
[abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % tree
.
├── backup
├── config
├── docs
├── library_info.txt
├── src
│   └── book.txt

5 directories, 2 files
[abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % █
```

Question 2 — Git Repository

```
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % nano library_info.txt
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   library_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git diff
diff --git a/library_info.txt b/library_info.txt
index 84cd214..1a94186 100644
--- a/library_info.txt
+++ b/library_info.txt
@@ -4,5 +4,7 @@ Subject: Application Build and Release Automation
Project Name: Online Library Management System
Semester: 5th
Project Objective: To develop an efficient Online Library Management System.
-Project Description: A system for managing and maintaining library information using Linux and Git.
+Project Description: A system for managing books, users, and library information using Linux and Git.
+.
```

■

```
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git add library_info.txt
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git commit -m "Update library information"
[main 4c274b7] Update library information
 1 file changed, 3 insertions(+), 1 deletion(-)
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git log --oneline
4c274b7 (HEAD -> main) Update library information
a03aac2 Initial project setup
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % █
```

Question 3 — Feature Branch

```
abhijectsingh@Abhijects-MacBook-Air Abhiject-GLA-LibrarySystem % git branch --show-current
book-feature
abhijectsingh@Abhijects-MacBook-Air Abhiject-GLA-LibrarySystem % git checkout main
Switched to branch 'main'
abhijectsingh@Abhijects-MacBook-Air Abhiject-GLA-LibrarySystem % git branch --show-current
main
abhijectsingh@Abhijects-MacBook-Air Abhiject-GLA-LibrarySystem % nano src/book.txt
abhijectsingh@Abhijects-MacBook-Air Abhiject-GLA-LibrarySystem % git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   src/book.txt

no changes added to commit (use "git add" and/or "git commit -a")
abhijectsingh@Abhijects-MacBook-Air Abhiject-GLA-LibrarySystem % git add src/book.txt
abhijectsingh@Abhijects-MacBook-Air Abhiject-GLA-LibrarySystem % git commit -m "Add book management module"
[main b9f1a98] Add book management module
 1 file changed, 1 insertion(+)
abhijectsingh@Abhijects-MacBook-Air Abhiject-GLA-LibrarySystem % git log --oneline
b9f1a98 (HEAD -> main) Add book management module
4c274b7 (book-feature) Update library information
a834ac2 Initial project setup
abhijectsingh@Abhijects-MacBook-Air Abhiject-GLA-LibrarySystem % █
```

Question 4 — Merge Conflict & Resolution

```
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   library_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % cat library_info.txt
Name: Abhijeet Singh
Roll Number: 2415001871
Subject: Application Build and Release Automation
<<<<<< HEAD
Project Name: Smart Online Library Management System
=====

>>>>>> library-update
Semester: 5th
Project Objective: To develop an efficient Online Library Management System.
Project Description: A system for managing books, users, and library information using Linux and Git.
.

abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % <<<<<< HEAD
Project Name: Smart Online Library Management System
=====
Project Name: Advanced Online Library Management System
>>>>>> library-update
zsh: parse error near '<<<'
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % nano library_info.txt
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git add library_info.txt
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git status
On branch main
All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

Changes to be committed:
  modified:   library_info.txt

abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git add library_info.txt
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git status
On branch main
All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

Changes to be committed:
  modified:   library_info.txt

abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git commit -m "Resolve library information merge conflict"
[main 8cc1c4] Resolve library information merge conflict
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % cat library_info.txt
Name: Abhijeet Singh
Roll Number: 2415001871
Subject: Application Build and Release Automation
Project Name: Smart Advanced Online Library Management System
Semester: 5th
Project Objective: To develop an efficient Online Library Management System.
Project Description: A system for managing books, users, and library information using Linux and Git.
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abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git status
On branch main
nothing to commit, working tree clean
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git log --oneline --all --graph
* 8cc1c4 (HEAD -> main) Resolve library information merge conflict
|
| * 8e7a2b9 (library-update) Update project name in library branch
| * | 2cf492a Update project name in main branch
|/
|
| * b9f1a98 Add book management module
| * 4c274b7 (book-feature) Update library information
| * a034ac2 Initial project setup
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % █
```

Question 5 — Final Submission

```
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abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % tree
.
├── backup
├── config
├── docs
├── library_info.txt
└── src
    └── book.txt

5 directories, 2 files
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git status
On branch main
nothing to commit, working tree clean
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git branch
book-feature
library-update
* main
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git log --oneline --all --graph
* 8cec1c4 (HEAD -> main) Resolve library information merge conflict
| \
| * 8e7a2b9 (library-update) Update project name in library branch
| * 2cf492a Update project name in main branch
|/
* b9f1a98 Add book management module
* 4c274b7 (book-feature) Update library information
* a834ac2 Initial project setup
abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % cat library_info.txt
Name: Abhijeet Singh
Roll Number: 2415081871
Subject: Application Build and Release Automation
Project Name: Smart Advanced Online Library Management System
Semester: 5th
Project Objective: To develop an efficient Online Library Management System.
Project Description: A system for managing books, users, and library information using Linux and Git.
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abhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % █
```